From Ultrasound Image Collection to De-identification and Re-identification: A Practical Pipeline

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Abstract: - Many AI research initiatives consider medical images a crucial resource to improve or enhance healthcare outcomes. The lack of high-resolution real-world image datasets, detailed annotations, and clinical relevance forces researchers to use public datasets as an alternative. The latter often impacts the accuracy of results and impedes further advancements of AI in this field. Meanwhile, in limited scenarios where researchers can collect real-world data, ensuring patient privacy becomes their primary concern. To minimize the risk of private information disclosure, images must be de-identified in a way that preserves their research value. Numerous studies focusing on de-identification approaches are available in the literature. However, there are often gaps or missing points in creating a real valuable dataset because simply de-identifying images is not sufficient. Creating medical image datasets for AI research projects involves many steps beyond just protecting patient identity. This study contributes to the existing research by presenting a comprehensive process for creating a clean and safe ultrasound images dataset, using real data as a basis. The authors introduce a real-world pipeline named UltraSafe, which serves as a semi-automated or automated tool that considers all the necessary steps, such as on-site ultrasound data collection from a private clinic, data cleaning, annotation, de-identification, and re-identification.

Key-Words: - Ultrasound images, Data collection, De-identification, Re-identification, Artificial intelligence, Protected health information.

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1 Introduction

Manual image segmentation, comparative analysis, classification, and interpretation are some of the routine, time-consuming tasks traditionally performed by radiologists. With the help of deep learning algorithms, which can effectively process large medical image datasets, AI can automate these tasks, [1]. Conversely, without an initial high-quality dataset, even the most efficient algorithms may underperform. Therefore, the performance and success of these algorithms depend significantly on the quality of the input data provided for training, [2].

Factors such as limited access, privacy concerns, limited computational resources, financial costs, and human errors pose challenges for building datasets based on real images. Consequently, researchers often adopt publicly available data in their studies, but they face many limitations and challenges associated with it, [3]. Many publicly available datasets are limited in size, making it almost impossible to find a high quantity of well-annotated cases suitable for training effective DL algorithms, [4]. Many authors use AI to annotate images, while others reduce their quality by decreasing the size or resolution. Additionally, some of the images included in these public datasets are screenshots from existing literature. All the actions mentioned previously can cause inaccuracies in any analysis conducted using these datasets, [3].

The existing datasets released through the years have significantly contributed to the development of DL algorithms that can effectively perform high-quality image analysis. Their success has been proven, but it is now necessary to refine and perfect

the process of creating a valuable dataset to achieve further advancements in AI rather than just the bare minimum. An important challenge that limits our progress is often the lack of connection between the data sourcing stage and its utilization, [5]. In most cases, the data sourcing process is performed with specific goals or objectives in mind and maintaining a particular data structure and quality, but different authors may have varying goals when leveraging the same set of data. This contrast will subsequently impact how the data will be analyzed later.

As a result, it takes the implementation of a standardized process to build a valuable medical image dataset. This article provides the "UltraSafe" roadmap that includes the whole process of collecting, cleaning, annotating, and making ultrasound images free from privacy concerns. In addition, the general context in which the dataset is created via this roadmap by the authors does not impact the context in which its data can be later used. To our knowledge, there are no published articles that provide practical insights or document the complete process of such an initiative. We have provided an entire pipeline that manages real US data through different stages. Unlike prior studies that use synthetic data, focus on other image modalities, or work on isolated components of the pipeline, our research validates the procedure in a real-world context. Moreover, the inclusion of the re-identification step to re-connect patients is often unexplored in medical image processing.

The remaining article has been arranged as follows. In Section II, we mention some results from the literature review. The methodology used in this study to create a valuable ultrasound dataset is portrayed in Section III. In section IV, we present the results and discussions on the pipeline that was followed. We mention the limitations faced by the authors and some future directions in this area, as well. Section V concludes the article by highlighting the major challenges faced during the process and providing further research suggestions.

2 Literature Review

The use of medical image data for research purposes is gaining more importance for improving the accuracy of deep learning tasks. As a result, building large-scale image datasets and secure data sharing for the secondary use of such data have become significant issues and topics for researchers in the field, [6], [7].

The authors in [8] built a DICOM dataset composed of various medical image modalities collected from existing datasets present in TCIA (The Cancer Imaging Archive) to assess the effectiveness of de-identification algorithms. In [9] was utilized a public dataset of CT images to perform facial de-identification. They created a modular solution to normalize such data and integrate it with publicly available archives like PACS and DICOM. The researchers in [10] presented a pipeline with various steps needed to prepare a CT scan dataset based on public well-known datasets to help authors interested in lung nodule research. This pipeline included collection, annotation, and preprocessing procedures, but the authors did not mention any de-identification step. The study by [11] contributed to the creation of a de-identified endoscopic dataset using public data sources. It can be used by researchers as a benchmarking tool to evaluate the performance of deep learning models for polyp classification.

In [12] the use of published was avoided. Instead, they created a de-identified X-ray dataset and made it publicly available. Although they contributed to the research by adding real anonymized images, they collected the data with a specific goal in mind: to enhance and improve the detection of congenital heart disease. Other researchers developed a new de-identification method for head CT images using face deformation [13]. The authors were focused on the process of privacy preservation while keeping intact the results of deep learning algorithms, and the technique was applied to a nonpublic dataset, as well.

The paper by [14] compared three different de-identification solutions that hide private information found in the voxels of 3D MRI scans. They concluded that despite the visual differences obtained by each evaluated method, all of them were reliable, even though a small bias might be present. On the other hand, other authors [15] concentrated more on the reverse process, known as re-identification. They introduced a deep learning system able to re-identify patients from their chest X-ray images and tested the system on a well-known publicly available dataset.

The work of [16] was more about the challenges regarding data storage and the advanced technology skills required to address them. Due to limited human and financial resources to offer such skills, medical image researchers choose outsourcing strategies as a solution. It is not only about saving the data for further use but storing it in a scalable way facilitating its access, sharing, and management. Cloud storage is often a solution that aligns well with such needs [17], [18]. The study by [19] highlighted the importance of proper data storage to establish a standardized workflow to

manage radiology images. Meanwhile, [20] proposed an open-source framework to stream images at different resolutions from only one high-resolution copy to reduce data storage inefficiencies.

Many researchers highlighted the importance of encrypting medical images with burned-in annotations or metadata coming along with them. The main reason for that is to prevent the exposure of PHI in image datasets used for e-health applications, research purposes, and medical AI pipelines for image classification or segmentation, [21], [22], [23]. Another reason is to preserve patient privacy during image transmission and storage, [24].

3 Methodology

The performance of medical AI systems and models relies heavily on well-defined and high-volume medical image datasets. Building such datasets requires significant effort in terms of time and costs. In addition, accessing and working with real medical image data necessitates compliance with well-known privacy legislation protecting patients' privacy, [25]. Moreover, clinics, hospitals, and other healthcare institutions rely on integrated systems to manage data. In such situations, ensuring compliance with GDPR and NIS2 becomes more difficult, [26].

A detailed step-by-step procedure is required to create a valuable dataset. Figure 1 (Appendix) introduces the “UltraSafe” pipeline followed by the authors to collect, clean, annotate, de-identify, and re-identify real-world data collected from the private “Multi Medical Center” clinic.

3.1 Data Collection

The images that we have taken into consideration in this study are in TIFF format and were collected via an external USB from an ACUSON X700 US system. All the images of the same patient come together with a StudyDB.xml file containing all the corresponding metadata. During this first round of data collection, we gathered a total of 4,665 images. Amongst these, only 2,005 were valid, providing a view of the internal body structure and information about the patient. Their creation dates vary from October 2019 to July 2024. The other 2,660 resulted in blank images because they were only printed for the patients but not saved in the system by the doctor. Given that, the latter were excluded from further processing.

It is worth noting that the StudyDB.xml files, besides the personal information about the patients, do not include details about the diagnosis for the specific visit during which the TIFF files were stored. All the patient's diagnoses and the related comments of the doctor are recorded in a separate external system named ‘eDoctor’.

3.2 Data Cleaning and Annotation

All the ultrasound images belonging to a patient follow the same filename format, 'timestamp_i.tif'. The 'timestamp' indicates the date and time the file was created, and 'i' ranges between 1 and 'n', with 'n' representing the number of files stored for that specific patient visit. It often happens that a visit takes place, and the doctor prints out the images for the patient without making sure that they get stored on the device, as well. This will result in empty images with blank or non-valid content. To address the issue, we start our data cleaning step with a basic Python script named 'check_image_validity.ipynb'. It distinguishes non-valid images from those having valid gynecological content. The distinction is performed based on the following criteria:

1. Non-valid images tend to occupy more space on the disk than images with content, exceeding 2 MB (specifically, 2.25 MB).
2. Non-valid images are all black and contain one of the following texts: "Report," "Early OB Report," "OB Report," "GYN Report," or "Fetal Echo Report."

Once all the blank images were removed and a folder containing only the valid images of the patients was created, we proceeded with the "generate_csv_with_filenames.ipynb" script. It consists of two steps:

First step: Generating a CSV file where each entry represents the filename of a valid image. This file, named 'us_img_filenames.csv', does not include a header.

Second step: Locating the corresponding StudyDB.xml file for each of the file names listed in the CSV and converting it into a JSON file. Once this step is completed for all entries, the resulting JSON files are joined into a common CSV file named 'patient_data.csv'. After cleaning the file and removing duplicates, the following columns remain: "DicomPatientID," "Patient.GivenName," "Patient.FamilyName," "Patient.CreationDate," "Patient.Sex," "Series.DicomProtocolName," and "Date of Birth." The "Age" column will be a derived attribute, calculated from the date of birth. For each record of the 'patient_data.csv', a match is found in the system that stores the diagnoses of patients. As a result, the columns "Code, Examination, and

Doctor's Comments" are added for each entry. This new file can be used to create in the future a reporting system that produces important KPIs for the clinic.

3.3 Data De-identification and Re-identification

The diagnostic and research value of medical images can be compromised by full image de-identification. Maintaining the balance between data utility and data privacy is crucial, [27]. Partial de-identification is often considered as a solution to minimize utility loss.

To proceed further with preparing the dataset to be used for secondary purposes, such as research, we need to revisit our 'valid_images' folder. Our goal is to determine whether the images follow a specific structure, which will allow us to create a generalized de-identification procedure. While manually checking the images, we noticed that information regarding the patient, clinic, and machine is only found in the top, left, and bottom areas of the image. This is illustrated in the example ultrasound image retrieved from the machine shown in Figure 2.

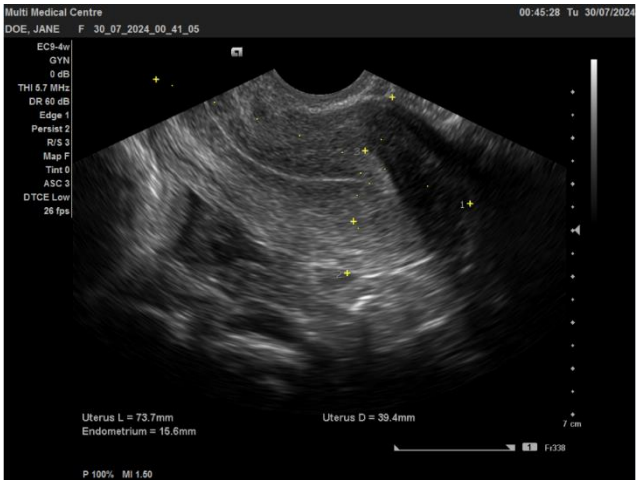

Fig. 2: An ultrasound image obtained from the machine

Given that, there is no point in de-identifying the whole image where no sensitive information is present in other areas. Hence, we have chosen to de-identify only specific regions of the image. These regions are outlined in the following lines of code.

```
region_T= img[0:int(0.09*h), :]
region_L= img[int(0.083*h):, 0:int(0.14*w)]
region_B= img[int(0.84*h):, :]
```

De-identification applied to the top area begins at the top edge (0) and extends down to 9% of the image height. For the left area, de-identification starts at the left edge (0) and covers 14% of the image width, beginning at 8.3% of the height from the top of the image. The bottom region extends up to 16% of the image height, starting from the bottom edge (0).

Various existing de-identification pipelines leverage the advantages of optical character recognition (OCR) to detect protected health information (PHI) in medical images. Authors in [28] adopted OCR in their pipeline to automate the process of detecting and removing burned-in annotations in ultrasound images without manual intervention. Similarly, in [29] OCR was integrated into the Pylogic library to extract and replace or remove PHI from ultrasound images, achieving a high success rate (average Dice Similarity Coefficient of 96%). Meanwhile, authors in [30] were mainly focused on object detection through DL. They used OCR and enhanced the robustness of the de-identification solution.

In this work, we blurred the regions containing sensitive information and used Tesseract OCR to extract metadata about the content and exact coordinates of the text present in each image. Afterward, we encrypted the metadata and covered the detected text areas with white rectangles. The de-identified image is depicted in Figure 3.

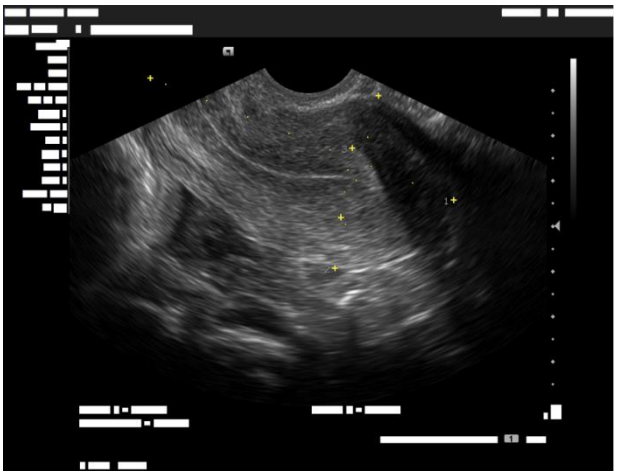

Fig. 3: De-identification of the original image

In various deep-learning tasks performed on medical images, such as segmentation or classification, it is essential to reconnect the image to the patient it belongs to at the end of the process. For that purpose, in our pipeline, the stored encrypted metadata gets decrypted and used to redraw the original image. The step of re-identification is illustrated in Figure 4.

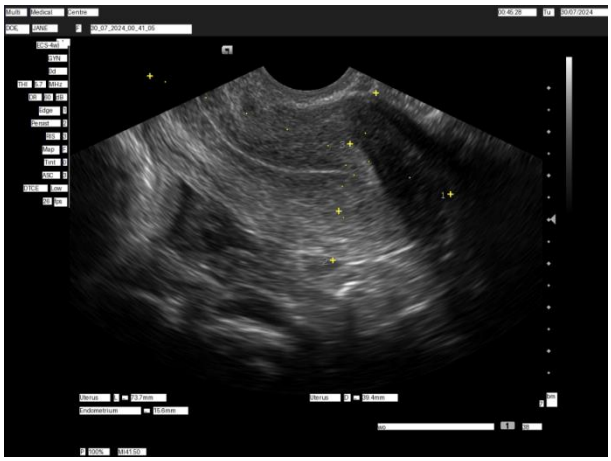

Fig. 4: Re-identification of the original image

4 Results and Discussions

UltraSafe pipeline serves as a semi-automated or automated tool that can be followed by other authors to collect ultrasound medical images from various institutions and build valuable medical image datasets. Following and implementing the UltraSafe pipeline has led the authors to the creation of two important outputs, for instance, a dataset of de-identified and re-identified ultrasound images and a dataset containing information about patients and their diagnoses.

The first output was produced by applying this [31] code on 4,665 images collected from the ACUSON X700 Ultrasound System owned by the private 'Multi Medical Center' clinic. From the whole collection, we de-identified only a total of 2005 images, which contained sensitive information and valid gynecological content. Subsequently, to evaluate the performance of de-identification, we executed the 'de-identification_validation.py' script for automated validation. Out of the images processed, 33 failed the validation, primarily due to the following reasons:

- The clinician incorrectly saved the ultrasound image, capturing more than just the image area and including unnecessary details from the entire machine interface. As a result, the coordinates required adjustment to properly focus on the correct part of the image.
- The clinician took a screenshot that displayed four different views of the patient within a single image. For that reason, the code must consider sensitive information located in various areas of the image.
- Text present in the bottom area was sometimes located above the considered 16% of the image height starting from the bottom edge.

Despite the unsuccessful cases, the automatic validation of de-identification on the removed Protected Health Information (PHI) resulted in a success rate of nearly 98%. Re-identification was also evaluated automatically using the 're-identification_validation.py' script, achieving a success rate of approximately 90%. This is due to minor text extraction errors or misreading characters. As suggested by the authors in [32], leveraging noise reduction techniques, such as adaptive median filtering, might enhance image quality and improve the re-identification success rate.

The calculated values for Precision (88.4%), Recall (98%), and F1-score (93%) were obtained by applying Eq. (1), (2), and (3).

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

This ultrasound image dataset can be further used by the authors to train deep-learning models performing ultrasound image classification and segmentation tasks.

Regarding the second output, a CSV file was created containing 373 records of patient data and their corresponding diagnoses. The data was filtered for PatientUID and CreationDate to remove duplicates. Over time, the size of this file will increase, and it might be used to generate health reports that indicate important performance and quality metrics for the clinic to improve its services.

4.1 Limitations and Future Directions

The pipeline is limited to images collected from the SIEMENS ACUSON x700 machine used in the private clinic that collaborated with us. Half of the collected images were blank images and failed validation since the clinicians did not store them properly. They need to be aware of the importance that proper data storage has in building a high-quality dataset. In our study, we use predefined coordinates that can be customized based on specific needs to perform PHI de-identification. For future work, we plan to explore deep learning-based adaptive approaches for detecting PHI. The authors in reference [33] highlight the success of Generative Adversarial Networks (GANs) in supporting robust classification tasks, so we will investigate GAN-based de-identification methods to address challenges related to image format variability. Also,

to improve the success rate of re-identification, we will consider using noise reduction techniques to enhance the quality of ultrasound images. In addition, data annotation is performed by two of the doctors working in the clinic. Even if the process is conducted carefully, some subjectivity may still occur.

Researchers can customize the pipeline to meet their specific needs. They may replicate the pipeline for different medical image modalities or expand the solution by building a dataset that includes mixed image modalities. They can also consider testing the workflow on ultrasound images collected from machines other than the SIEMENS ACUSON x700.

To make this research more applicable, the authors aim to use the dataset obtained from this pipeline to perform deep learning tasks such as image segmentation and classification.

5 Conclusion

Due to many challenges that limit practitioners' ability to create high-quality datasets from real images, they often opt for publicly available datasets for their deep learning projects. Besides that, in the literature, we were not able to find a study where authors present a detailed, step-by-step procedure showing what it takes to build a good medical image dataset. This workflow should encompass all the phases, from data collection in the selected institution to the preparation of the dataset for subsequent use in classification, segmentation, or other deep learning tasks. Since medical images are considered sensitive in terms of privacy, this procedure must include de-identification and re-identification of images, as well.

Given that, we decided to build our ultrasound image dataset. For that, we created the 'UltraSafe' pipeline, which serves as a framework that can be followed, customized, and, at the same time, improved by other researchers based on their needs. In this roadmap, we outline the process of collecting ultrasound images from the private "Multi Medical Center" clinic. We gathered a total of 4,665 images, including valid and non-valid ones, stored from October 2019 to July 2024. After that, we performed data cleaning by initially removing all the non-valid images remaining with a total of 2,660 valid ultrasounds. Consequently, we processed these images and created a CSV file with all the patient data corresponding to each image. Furthermore, we connected them with the diagnoses and doctors' comments stored in a separate system. The final CSV file was de-identified through partial de-identification of patients' protected health information (PHI). The regions containing such information were blurred, and image metadata was extracted through tesseract OCR. As a final step, the metadata was encrypted and used later for the process of re-identification to reconnect the image to the patient it belongs.

Finally, the authors mention some of the limitations faced and future directions for themselves, practitioners, or other researchers aiming to improve and expand this pipeline.

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The authors have no conflicts of interest to declare.

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APPENDIX

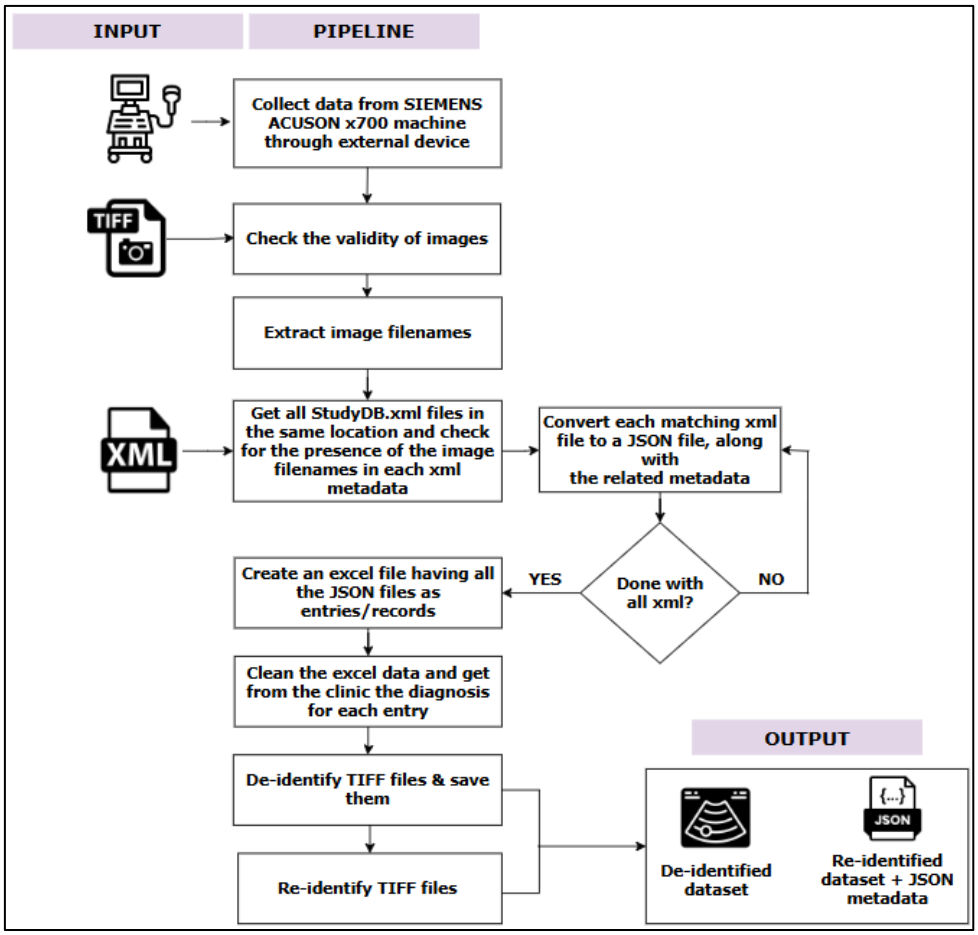

Fig. 1: UltraSafe Pipeline